

Lesson 16 — Build the Model & Run a Sequence

Matches LEGO Teacher Portal lesson: "Nest Quest".

Objective (I can...): I can build the sequences model and run an ordered set of steps that finishes a job.

CS concept: Sequences (ordered steps that work together to do a job) · **Builds on:** Lesson 15 (events start a program; now we grow the ordered steps after the start).

AI Connection (2 min, read aloud): A voice helper follows a sequence every time: wake → listen → think → answer. The order is always the same. *Ask the group:* "What would go wrong if it answered before it listened?"

Materials: LEGO CS & AI kit + device per pair, 4-5 big "step" picture cards per pair. **This is the unit's build lesson** — build the Sequences model now and keep it assembled through Lesson 20.

60-minute flow - Warm-up (10): "Morning march" — pairs put 5 "getting ready" picture cards in order (wake, dress, eat, shoes, go) and march through them. A sequence is steps in the right order. - **Concept + demo (8):** Show a 4-5 step program running top to bottom (light → motor spins → wait → beep → stop). Point to each step as it runs. The whole thing is a sequence. - **Build/Code (25):** Pairs **build the Sequences model** (the unit's one build), then code a 4-5 step sequence that finishes a little job, running it to check each step happens in order. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add one more step in the right spot; run and point to where it fits. - **Share + tidy-up (10):** Pairs run their sequence and count the steps out loud. **Leave the model built.**

Pair roles: Driver codes, Navigator points to the step cards and counts along. Swap at the wiggle break. (*Trio rotates.*)

Differentiation: *Support* — a 3-step sequence with the first step placed. *Extension / fast-finisher* — a 6-step sequence. *Reteach* — run one step at a time and name it before adding the next.

Exit check: Pair runs a 4+ step sequence in the correct order. **At-home (optional):** Draw a 5-picture sequence for making a snack, in order. ~5 min.
